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JEE MAIN PHYSICS DPP: INCLINED PROJECTILES & UCM

Topic: Projectile on Inclined Plane & Uniform Circular Motion (Angular Kinematics & Centripetal Acceleration)

| Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

Q1. A particle is projected from the base of an inclined plane of inclination β with initial velocity u at an angle α relative to the inclined surface. The time of flight T is:

- (a) $\frac{2u \sin \alpha}{g \sin \beta}$
 (b) $\frac{2u \sin \alpha}{g \cos \beta}$
 (c) $\frac{2u \cos \alpha}{g \cos \beta}$
 (d) $\frac{u \sin \alpha}{g \cos \beta}$

Q2. For a given projection speed u up an inclined plane of inclination β , the angle of projection α with the incline that maximizes the range up the plane is:

- (a) $\frac{\pi}{4}$
 (b) $\frac{\pi}{4} + \frac{\beta}{2}$
 (c) $\frac{\pi}{4} - \frac{\beta}{2}$
 (d) $\frac{\pi}{2} - \beta$

Q3. A particle is projected up an incline of angle $\beta = 30^\circ$ with speed u . If it strikes the inclined plane perpendicularly, the angle of projection α with respect to the incline satisfies:

- (a) $\tan \alpha = \sqrt{3}$
 (b) $\tan \alpha = \frac{\sqrt{3}}{2}$
 (c) $\tan \alpha = 2\sqrt{3}$
 (d) $\tan \alpha = \frac{\sqrt{3}}{6}$

Q4. The maximum range $R_{\max}$ of a projectile fired down an inclined plane of inclination β with speed u is:

- (a) $\frac{u^2}{g(1 - \sin \beta)}$
 (b) $\frac{u^2}{g(1 + \sin \beta)}$
 (c) $\frac{u^2 \cos \beta}{g(1 - \sin \beta)}$
 (d) $\frac{u^2}{g \cos^2 \beta}$

Q5. A particle moves in a circle of radius $r = 0.5$ m with a constant angular speed $\omega = 4$ rad/s. The linear speed v and centripetal acceleration a_c of the particle are:

- (a) $v = 1$ m/s, $a_c = 4$ m/s²
 (b) $v = 2$ m/s, $a_c = 8$ m/s²
 (c) $v = 2$ m/s, $a_c = 4$ m/s²
 (d) $v = 4$ m/s, $a_c = 16$ m/s²

Q6. A particle moves in uniform circular motion of radius R . If its time period of revolution is T , the magnitude of average acceleration during half a revolution is:

- (a) $\frac{2\pi R}{T^2}$
 (b) $\frac{4\pi R}{T^2}$
 (c) $\frac{8\pi R}{T^2}$
 (d) 0

Q7. For a particle in uniform circular motion, the position vector $\vec{r}$ and centripetal acceleration vector $\vec{a}_c$ satisfy which relation?

- (a) $\vec{a}_c \cdot \vec{r} = 0$
 (b) $\vec{a}_c \times \vec{r} \neq \vec{0}$
 (c) $\vec{a}_c = \omega^2 \vec{r}$
 (d) $\vec{a}_c = -\omega^2 \vec{r}$

Q8. A particle of mass m is executing uniform circular motion of radius R with constant speed v . The change in momentum of the particle in traversing an arc subtending an angle θ at the center is:

- (a) $2mv \sin\left(\frac{\theta}{2}\right)$
 (b) $2mv \cos\left(\frac{\theta}{2}\right)$
 (c) $mv \sin \theta$
 (d) $mv(1 - \cos \theta)$

Q9. The minute hand of a clock has a length of 6 cm. The magnitude of the displacement vector and average velocity of the tip of the minute hand from 12 : 00 PM to 12 : 30 PM are:

- (a) 6 cm, $\frac{1}{300}$ cm/s
 (b) 12 cm, $\frac{1}{150}$ cm/s
 (c) 12 cm, $\frac{\pi}{150}$ cm/s
 (d) 0 cm, 0 cm/s

- Q10.** A body is projected from an incline of angle β with velocity u at angle α to the incline. The maximum height $H_{\max}$ reached above the inclined surface is:
- $\frac{u^2 \sin^2 \alpha}{2g}$
 - $\frac{u^2 \sin^2(\alpha + \beta)}{2g}$
 - $\frac{u^2 \sin^2 \alpha}{2g \cos \beta}$
 - $\frac{u^2 \cos^2 \alpha}{2g \sin \beta}$
- Q11.** If a body is executing uniform circular motion, which of the following quantities remains strictly constant?
- Velocity vector
 - Acceleration vector
 - Momentum vector
 - Kinetic energy
- Q12.** A particle moves in a circle of radius 20 cm at 120 rpm. The magnitude of its centripetal acceleration is: (Take $\pi^2 \approx 9.87$)
- $3.2\pi^2 \text{ m/s}^2 \approx 31.58 \text{ m/s}^2$
 - $1.6\pi^2 \text{ m/s}^2$
 - $0.8\pi^2 \text{ m/s}^2$
 - $6.4\pi^2 \text{ m/s}^2$
- Q13.** Two particles P and Q move in concentric circles of radii R_P and R_Q with the same time period T . The ratio of their centripetal accelerations $\frac{a_P}{a_Q}$ is:
- $\frac{R_Q}{R_P}$
 - $\frac{R_P}{R_Q}$
 - $\frac{R_P^2}{R_Q^2}$
 - 1
- Q14.** A particle is projected from the ground with speed u up an inclined plane of inclination β . The ratio of the maximum range up the plane to the maximum range down the plane for the same launch speed u is:
- $\frac{1 - \sin \beta}{1 + \sin \beta}$
 - 1
 - $\frac{1 - \sin \beta}{1 + \sin \beta}$
 - $\frac{\cos \beta}{1 + \sin \beta}$
- Q15.** An object moves along a circular path of radius R in uniform circular motion with linear momentum p . The centripetal force acting on the object is given by:

- $p^2 R$
- $\frac{p}{R}$
- $p\omega^2$
- $\frac{p^2}{mR}$

- Q16.** A particle projected from the base of an incline of angle β lands at the same level of the incline after time T . The horizontal component of displacement is given by:

- $R \cos \beta$
- $R \sin \beta$
- $R \tan \beta$
- $R \sec \beta$

- Q17.** If angular displacement of a rotating wheel varies with time as $\theta(t) = 2t^3 - 3t^2$ rad, the angular acceleration at $t = 2$ s is:

- 12 rad/s^2
- 18 rad/s^2
- 6 rad/s^2
- 24 rad/s^2

- Q18.** A wheel of radius $R = 0.4$ m rotates with constant angular speed $\omega = 10$ rad/s. The distance traveled by a point on the rim in 5 seconds is:

- 10 m
- 15 m
- 20 m
- 25 m

- Q19.** For an inclined plane of inclination β , a projectile is thrown up the plane. If the initial velocity is purely perpendicular to the incline ($\alpha = 90^\circ$), the time of flight T is:

- $\frac{u}{g \cos \beta}$
- $\frac{2u}{g \sin \beta}$
- $\frac{u}{g \sin \beta}$
- $\frac{2u}{g \cos \beta}$

- Q20.** A projectile is launched down an inclined plane of angle 30° at an angle of 30° to the incline with speed $u = 10$ m/s. The range along the incline is: (Take $g = 10 \text{ m/s}^2$)

- $\frac{40}{3} \text{ m} \approx 13.33 \text{ m}$
- 10 m
- 20 m
- $\frac{20}{3} \text{ m}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

Setting axis perpendicular to the incline ($a_{y'} = -g \cos \beta$, $u_{y'} = u \sin \alpha$):

$$y'(T) = (u \sin \alpha)T - \frac{1}{2}(g \cos \beta)T^2 = 0$$

$$T = \frac{2u \sin \alpha}{g \cos \beta}$$

Sol 2. Correct Option: (c)

Range up the incline is:

$$R = \frac{u^2}{g \cos^2 \beta} [\sin(2\alpha + \beta) - \sin \beta]$$

R is maximized when $\sin(2\alpha + \beta) = 1$:

$$2\alpha + \beta = \frac{\pi}{2} \implies \alpha = \frac{\pi}{4} - \frac{\beta}{2}$$

Sol 3. Correct Option: (d)

For the projectile to hit the incline perpendicularly, its velocity component parallel to the incline must be zero at landing ($v_{x'} = 0$):

$$v_{x'} = u \cos \alpha - (g \sin \beta)T = 0$$

Substituting $T = \frac{2u \sin \alpha}{g \cos \beta}$:

$$u \cos \alpha = 2u \sin \alpha \tan \beta$$

$$\tan \alpha = \frac{\cot \beta}{2} = \frac{\sqrt{3}}{2}$$

Sol 4. Correct Option: (a)

For projection down an incline of angle β :

$$R = \frac{u^2}{g \cos^2 \beta} [\sin(2\alpha - \beta) + \sin \beta]$$

Maximized when $\sin(2\alpha - \beta) = 1$:

$$R_{\max} = \frac{u^2(1 + \sin \beta)}{g(1 - \sin^2 \beta)} = \frac{u^2}{g(1 - \sin \beta)}$$

Sol 5. Correct Option: (b)

- Linear speed: $v = r\omega = 0.5 \times 4 = 2$ m/s.
- Centripetal acceleration: $a_c = \omega^2 r = 4^2 \times 0.5 = 8$ m/s².

Sol 6. Correct Option: (c)

During half a revolution ($\Delta t = T/2$), velocity vector reverses from $\vec{v}$ to $-\vec{v}$:

$$|\Delta \vec{v}| = 2v = 2 \left(\frac{2\pi R}{T} \right) = \frac{4\pi R}{T}$$

Average acceleration:

$$a_{\text{avg}} = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{4\pi R/T}{T/2} = \frac{8\pi R}{T^2}$$

Sol 7. Correct Option: (d)

In uniform circular motion, the acceleration is directed toward the center, opposite to the radial position vector $\vec{r}$:

$$\vec{a}_c = -\omega^2 \vec{r}$$

Sol 8. Correct Option: (a)

$$|\Delta \vec{p}| = \sqrt{2p^2(1 - \cos \theta)} = 2p \sin(\theta/2)$$

Magnitude: $|\Delta \vec{p}| = 2mv \sin(\theta/2)$.

Sol 9. Correct Option: (b)

In 30 minutes, the minute hand sweeps 180°:

- Displacement: $d = 2 \times 6 = 12$ cm.
- Time: $\Delta t = 30 \times 60 = 1800$ s.

Average velocity:

$$v_{\text{avg}} = \frac{12 \text{ cm}}{1800 \text{ s}} = \frac{1}{150} \text{ cm/s}$$

Sol 10. Correct Option: (c)

Using $v_{y'}^2 = u_{y'}^2 - 2a_{y'}H_{\max}$ with $v_{y'} = 0$ at peak:

$$0 = (u \sin \alpha)^2 - 2(g \cos \beta)H_{\max}$$

$$H_{\max} = \frac{u^2 \sin^2 \alpha}{2g \cos \beta}$$

Sol 11. Correct Option: (d)

In UCM, speed v is constant, so kinetic energy $K = \frac{1}{2}mv^2$ is strictly constant. Direction-dependent vectors ($\vec{v}$, $\vec{a}$, $\vec{p}$) continuously change.

Sol 12. Correct Option: (a)

$$\omega = 120 \text{ rpm} = 120 \times \frac{2\pi}{60} = 4\pi \text{ rad/s}$$

Radius $r = 0.2$ m.

$$a_c = \omega^2 r = (4\pi)^2 \times 0.2 = 16\pi^2 \times 0.2 = 3.2\pi^2 \text{ m/s}^2$$

Sol 13. Correct Option: (b)Centripetal acceleration in terms of period: $a_c =$

$$\omega^2 R = \left(\frac{2\pi}{T}\right)^2 R.$$

Since T is identical:

$$\frac{a_P}{a_Q} = \frac{R_P}{R_Q}$$

Sol 14. Correct Option: (c)

$$\bullet \text{ Up the plane: } R_{\max, \text{ up}} = \frac{u^2}{g(1 + \sin \beta)}.$$

$$\bullet \text{ Down the plane: } R_{\max, \text{ down}} = \frac{u^2}{g(1 - \sin \beta)}.$$

Ratio:

$$\frac{R_{\max, \text{ up}}}{R_{\max, \text{ down}}} = \frac{1 - \sin \beta}{1 + \sin \beta}$$

Sol 15. Correct Option: (d)

$$p = mv \implies v = \frac{p}{m}.$$

Centripetal force:

$$F_c = \frac{mv^2}{R} = \frac{m(p/m)^2}{R} = \frac{p^2}{mR}$$

Sol 16. Correct Option: (a)

The range R is along the inclined surface (at angle β to horizontal). Resolving horizontally gives $x = R \cos \beta$.

Sol 17. Correct Option: (b)

$$\omega(t) = \frac{d\theta}{dt} = 6t^2 - 6t.$$

$$\alpha(t) = \frac{d\omega}{dt} = 12t - 6.$$

At $t = 2$ s:

$$\alpha(2) = 12(2) - 6 = 18 \text{ rad/s}^2$$

Sol 18. Correct Option: (c)Linear speed $v = \omega R = 10 \times 0.4 = 4$ m/s.Distance traveled in $t = 5$ s:

$$s = v \times t = 4 \times 5 = 20 \text{ m}$$

Sol 19. Correct Option: (d)

$$T = \frac{2u \sin \alpha}{g \cos \beta}. \text{ For } \alpha = 90^\circ (\sin 90^\circ = 1):$$

$$T = \frac{2u}{g \cos \beta}$$

Sol 20. Correct Option: (a)For projection down an incline ($\beta = 30^\circ$, $\alpha = 30^\circ$):

$$u_{x'} = 10 \cos 30^\circ = 5\sqrt{3} \text{ m/s}, \quad u_{y'} = 10 \sin 30^\circ = 5 \text{ m/s}.$$

$$a_{y'} = -g \cos 30^\circ = -5\sqrt{3} \text{ m/s}^2, \quad a_{x'} = +g \sin 30^\circ = +5 \text{ m/s}^2.$$

Time of flight:

$$T = \frac{2(5)}{5\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ s}$$

Range down the incline:

$$R = u_{x'} T + \frac{1}{2} a_{x'} T^2 = 5\sqrt{3} \left(\frac{2}{\sqrt{3}}\right) + \frac{1}{2}(5) \left(\frac{4}{3}\right)$$

$$R = 10 + \frac{10}{3} = \frac{40}{3} \text{ m}$$

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